Advanced Combustion via Microgravity Experiments (ACME)

PMT Characterization Report

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PREFACE

Under the Space Flight Systems Development and Operations Contract (SpaceDOC), the National Aeronautics and Space Administration (NASA) has developed a modular, multi-user experimentation facility for conducting fluid physics and combustion science experiments in the microgravity environment of the International Space Station (ISS). This facility, called the Fluids and Combustion Facility (FCF), consists of two test platforms: the Fluids Integrated Rack (FIR) and the Combustion Integrated Rack (CIR). Also included in SpaceDOC are the required support efforts for Mission Integration and Operations, consisting of the Telescience Support Center (TSC) and Mission Integration and Planning (MIP).

The Advanced Combustion via Microgravity Experiments (ACME) is mini-facility designed to support Principal Investigators (PI) interested in studying various aspects of advanced combustion in a microgravity environment. ACME will be integrated into the Combustion Integrated Rack (CIR) onboard the International Space Station in the US Lab Module.

ACME will perform microgravity experiments to study fire suppression/flame extinguishment via 5 experiments: s-flame, Flame design, CLD Flame, E-Field Flames, and Burning Rate Emulator Flames as described in the ACME Baseline Description Document, ACME-DOC-004. ACME will allow investigation using pure as well as mixtures of gaseous fuels at various chamber atmosphere pressures, compositions, and oxygen concentrations. The microgravity environment allows for observation and measurement of flame formation and extinguishment processes that cannot be made in sustained gravity. The modular design of ACME allows for flexibility in configuration for specific experiments and easy maintainability of the hardware.

The core elements of gaseous combustion are included in the ACME design as a basis for current and future gaseous combustion studies. These elements are establishment of gas flow through various exchangeable mass flow controllers, ignition of gas at the burner and the use of appropriate diagnostic measurement devices. The five identified experiments, and any future experiments, use these core elements to build a unique set of requirements for the study of various phenomena distinctive to gaseous combustion. The goal of ACME is to improve the fundamental understanding of combustion processes in order to enable increased efficiency and reduce pollutant emission in practical combustion.

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1.0 INTRODUCTION

1.1 Purpose

The purpose of this report is to document the results for the characterization of the ACME PMT (Photomultiplier Tube) Assembly (S1101MFA1500), Flight unit, S/N 1002.

1.2 Scope

The scope of this report covers all characterizations, calibrations and tests that were performed in March, 2016 on the ACME flight model. Results and subsequent analysis will be recorded in this final characterization report. This document does not cover all verification test results, but may be used as a reference for verifications.

1.3 Classification

The classification for the ACME PMT Assembly is Flight Model, including all the hardware listed in Table II.

2.0 DOCUMENTS

This section lists specifications, models, standards, guidelines, handbooks, and other special publications. These documents have been grouped into two categories: applicable documents and reference documents.

2.1 Order of Precedence for Documents

In the event of a conflict between this document and other documents referenced herein, the information of this document shall apply. In the event of a conflict between this document and the contract, the contractual requirements shall take precedence over this document. All documents used, applicable or referenced are to be the issues defined in the Configuration Management (CM) contract baseline. All document changes, issued after baseline establishment, shall be reviewed for impact on scope of work. If a change to an applicable document is determined to be effective and contractually approved for implementation, the revision status will be updated in the CM contract baseline. The contract revision status of all applicable documents is available by accessing the CM database. Nothing in this document supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.2 Applicable Documents

The following documents are applicable to the ACME project to the extent specified herein.

S1101MFA1500	ACME PMT Assembly
ACME-REQ-001 Rev. K	ACME Engineering Requirements Document
ACME-TST-031 Rev. A	Upstream Relay Connector Board Test Procedure
P13001	SpaceDOC Procedure for the Control of Nonconforming Product
P05002	Configuration Management Procedure
P30020	SpaceDOC Safety and Mission Assurance Plan

2.3 Reference Documents

The documents/drawings in this section are provided only as reference material for background information and are not imposed as requirements.

ANSI-ESD S20.20.2007	ESD Association for the Development of an Electrostatic Discharge Program for Protection of Electronic Parts, Assemblies and Equipment (excluding Electrically Initiated Explosive Devices)
ACME-DOC-002 Rev. D	ACME Compliance Matrix
ACME-DOC-007 Rev. A	ACME Concept of Operations
ACME-SRD-001 Rev. -	Integrated Science Requirements Document

3.0 FACILITIES, EQUIPMENT, AND INSTRUMENTATION

3.1 Test Facilities

The ACME flight PMT Assembly Characterization took place in March, 2016 in the ZIN Technologies lab facility located at 6749 Engle Road.

3.2 Equipment and Instrumentation

The following table lists the name and type of the equipment and instrumentation required for the ACME Data Camera Package Characterization. The calibration status of all instrumentation was verified as appropriate before the characterization began.

TABLE I. Equipment and Instrumentation Information

Equipment	Manufacturer	Model Number	Cal Due Date
Dual DC Power Supply, +3.3 VDC / 24 VDC	Tektronix	Model: PS-280 S/N: TW52908 NASA Tag: 12129	N/A (via voltmeter)
DC Regulated Power Supply, 28 VDC	Kenwood	Model: PD36-20AD S/N: 5010016 ZIN Tag: Z000162	N/A (via voltmeter)
Digital Voltmeter/Multi-meter	Fluke	Model: 179 S/N: 81730258 ZIN Tag: Z000371	11 May, 2016
Upstream Relay Connector Board Assembly	ZIN Technologies	S1101EFA0302 S/N: 1001	N/A
Upstream Relay Board Power Supply Test Cable	ZIN Technologies	S1101EGA4541 S/N: 1001	N/A
Upstream Relay Input Signal Test Cable	ZIN Technologies	S1101EGA4538 S/N: 1001	N/A
Upstream Relay Board I/O Signal Test Kit	ZIN Technologies	S1101EGA4543 S/N: 1001	N/A
ACME Benchtop Cube Assembly with UEI 5-layer enclosure containing AI-207, AI-255 & AO-333 adapters	ZIN Technologies	N/A	N/A
Laptop w/ Ethernet & Serial Port	Dell	Model: Latitude E5510 S/N: CR8W4N1 ZIN Tag: Z004622	N/A
Oscilloscope	Tektronix	Model: HPLS-36APL750 S/N: 0609520 NASA Tag: 43895	N/A
Blackbody Infrared Source	Infrared Systems Development	Model: IR-564 S/N: 3062K NASA Tag: 09083	7/2016
Blackbody Temp. Controller	Infrared Systems Development	Model: MSO-3014 S/N: C025276 ZIN Tag: 58723	4/2016

4.0 DESCRIPTION OF TEST ARTICLE & CONFIGURATION

4.1 Test Article

4.1.1 ACME PMT Assembly

The ACME PMT Assembly (S1101MFA1500) consists of three independent commercial Photomultiplier Tube (PMT) modules (Hamamatsu H10722-110) in a single harness assembly (PMT Harness Assembly, S1101EFA1512). This Harness Assembly is mounted on a bracket (PMT Bracket Assembly, S1101MFA1501), along with individual filters associated with each of the 3 PMTs:

- PMT Filter Housing Assembly 1 (OH*), S1101MFA1504
- PMT Filter Housing Assembly 2 (Broadband), S1101MFA1505
- PMT Filter Housing Assembly 3 (CH*), S1101MFA1506

Together these make up the entire PMT Assembly.

The PMT Assembly flight unit was assembled in-house by ZIN Technologies at 6749 Engle Road.

TABLE II. Test Article

Description	Manufacturer	Drawing Number	S/N
ACME PMT Assembly	N/A	S1101MFA1500	1002

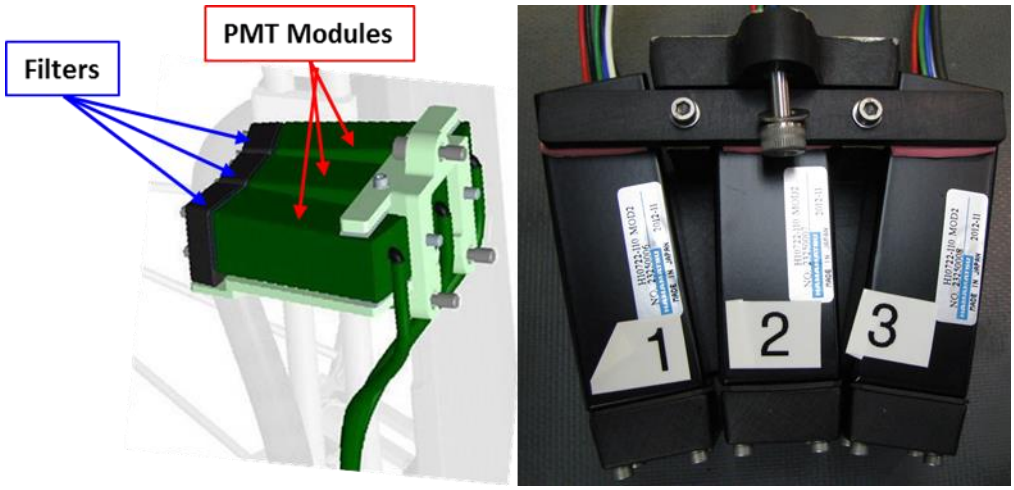


FIGURE 1. ACME PMT Assembly

4.1.1.1 Photomultiplier Tube (PMT) Module

Each Photomultiplier Tube (PMT) Module is a commercial product made by Hamamatsu. The module's metal enclosure contains the Photomultiplier Tube itself along with a low power consumption, high voltage power supply circuit and a low noise amplifier. The amplifier converts the PMT current output into a DC Voltage output signal. See a schematic of the PMT Module below.

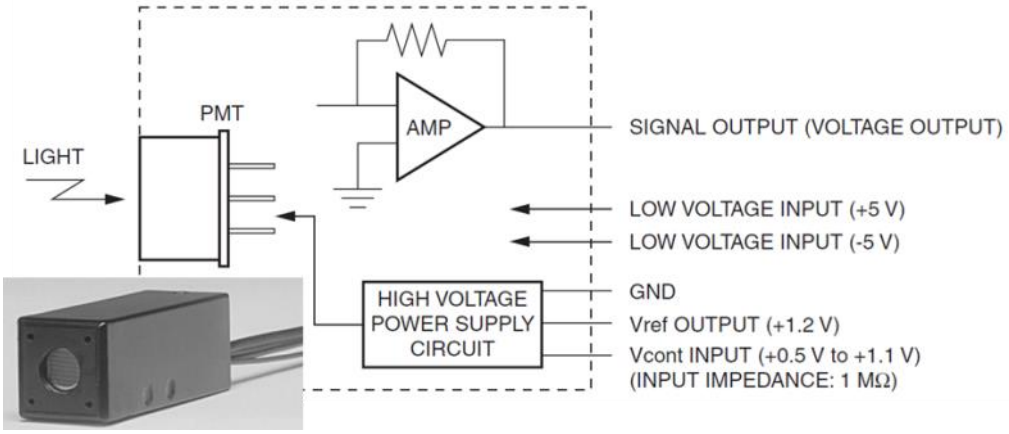


FIGURE 2. Hamamatsu PMT Module Schematic

The Hamamatsu model used for ACME is the H10722-110 PMT Module that is based on a Super Bi-Alkali photocathode with a spectral response from 230 nm to 700 nm (Hamamatsu Model R9880U-110). The PMT Module specifications are listed below.

TABLE III. Hamamatsu H10722-110 PMT Module Specs

Parameter			Rating	Unit
Frequency Bandwidth			DC to 20 kHz	-
Current to Voltage Conversion Factor			1 V/uA	-
Input Voltage			±4.5 to ±5.0	V
Max.Input Voltage			±5.5	V
Max.Input Current*1			+6.2/-3.5	mA
Max.Output Signal Voltage*8			+4 (Load resistance 10 kohm)	V
Max.Control Voltage			+ 1.15 (Input Impedance 1 Mohm)	V
Recommended Control Voltage Adjutment Range			+0.5 to +1.1	V
Effective Photocathode Area			8	mm dia.
Sensitivity Adjustment Range			$1:10^4$	-
Peak Sensitivity Wavelength			390	nm
Cathode	Luminous Sensitivity	Typ.	135	µA/lm
Cathode	Blue Sensitivity Index (CS 5-58)	Typ.	14	-
Cathode	Red/White Ratio	Typ.	-	-
Cathode	Radiant Sensitivity*4	Typ.	120	mA/W
Anode	Luminous Sensitivity*5	Typ.	$2.7x10^8$	V/lm
Anode	Radiant Sensitivity*4*5	Typ.	240	V/nW
Anode	Voltage Output Depending on	Typ.	1	mV
Anode	PMT Dark Current*5*6	Max.	10	mV
Offset Voltage		Typ.	±3	V
Ripple Noise*5*7 (peak to peak)		Max.	0.8	mV
Settling Time*2		Max.	10	s
Operating Ambient Temperature*3			+5 to +50	°C
Storage Temperature*3			-20 to +50	°C
Weight			100	g

4.1.1.2 PMT Gain

The PMT Gain is a direct function of the control voltage. This means that even when the input from a source is constant, if the control voltage increases, the PMT output will increase as well. Therefore the PMT Gain, via adjusting the input control voltage, is commonly used when the overall strength of the incoming signal varies. Adjusting the Gain appropriately allows the PMT to accommodate both strong and weak sources. The Gain vs. Control Voltage curve from Hamamatsu that is common to all of the ACME PMTs is shown below.

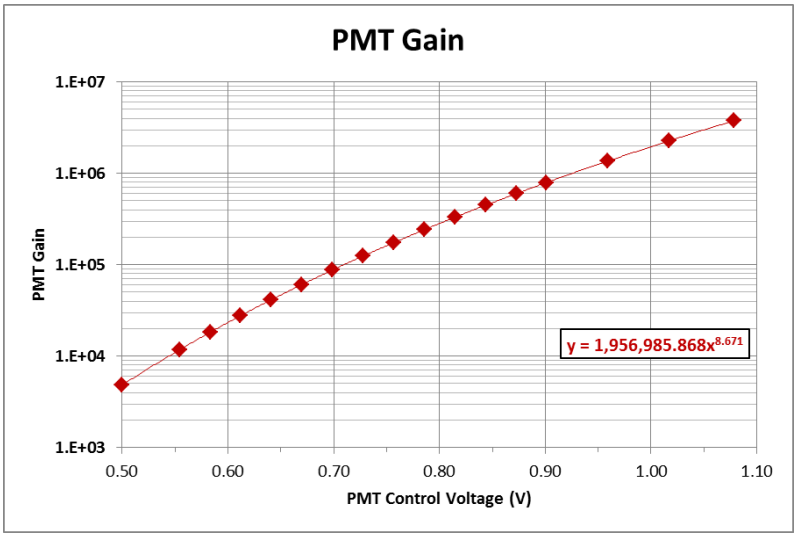


FIGURE 3. PMT Gain vs. Control Voltage

In addition to the imbedded electronics, ACME uses a Relay Board to convert the supply voltage range (0 – 10 VDC) from the Avionics Package into the specified usable PMT Control Voltage range (0.5 – 1.1 VDC). The Relay Board conversion to Control Voltage relationship is shown below.

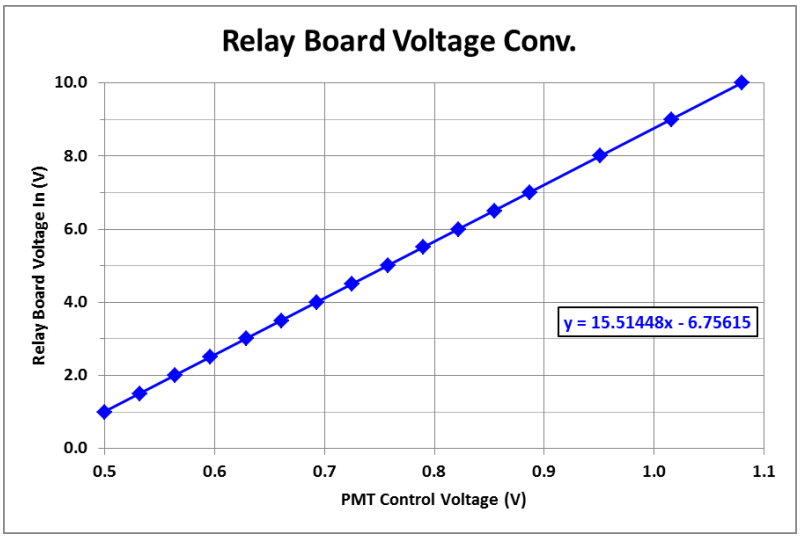


FIGURE 4. ACME Relay Board – PMT Voltage Conversion

The Table below shows a numerical representation of the Voltage conversion vs. Gain for the ACME PMTs.

TABLE IV. Voltage Conversion and Gain

Relay Board Input Voltage	PMT Control Voltage	PMT Gain
1.0	0.50	4.8E+03
1.5	0.53	8.2E+03
2.0	0.56	1.4E+04
2.5	0.60	2.2E+04
3.0	0.63	3.5E+04
3.5	0.66	5.4E+04
4.0	0.69	8.2E+04
4.5	0.73	1.2E+05
5.0	0.76	1.8E+05
5.5	0.79	2.5E+05
6.0	0.82	3.6E+05
6.5	0.85	5.0E+05
7.0	0.89	6.9E+05
8.0	0.95	1.3E+06
9.0	1.02	2.2E+06
10.0	1.08	3.8E+06

4.1.1.3 PMT Quantum Efficiency (QE)

The Quantum Efficiency (QE) of the Photomultiplier Tube (PMT) Module is the primary way to express the PMT sensitivity. QE is the conversion ratio of photoelectrons per photon (i.e. the number of photoelectrons emitted divided by the number of incident photons striking the photocathode), typically expressed as a percentage that varies with wavelength. It is critical to have the QE match up with the desired wavelengths to be measured. The H10722 series PMT Module QE is shown below.

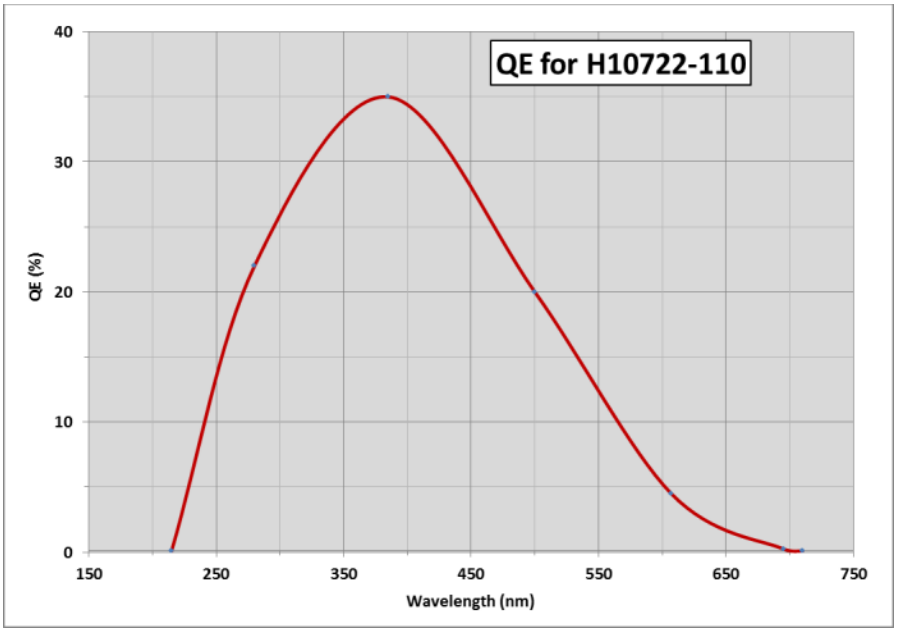


FIGURE 5. Hamamatsu H10722 PMT Quantum Efficiency (QE)

4.1.1.4 Filter Assembly 1 – OH* Filter

Filter Assembly 1 associated with PMT Module 1 contains a custom Narrow Bandpass Filter, provided by Alluxa, Inc. This filter has a Fused Silica substrate with a Narrow Bandpass coating. The coating center wavelength is ~ 310 nm with a FWHM (Full Width at Half Max) of just under 15 nm. See the filter curve and data below.

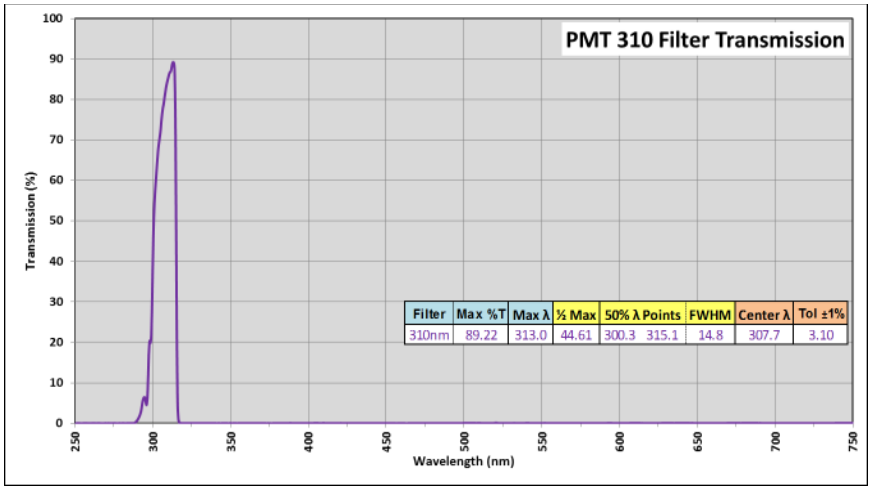


FIGURE 6. OH* Filter Curve

4.1.1.5 Filter Assembly 2 – ND 1.0 Filter

Filter Assembly 2 associated with PMT Module 2 contains a custom Neutral Density (ND) Filter, provided by Esco, Inc. This filter has a Fused Silica substrate with a broadband neutral density coating in order to attenuate the incoming light. The optical density of this coating is OD = 1.0, which corresponds to 10% transmission. See the filter curve and data below, along with the nominal transmission curve for Fused Silica before the ND coating is applied.

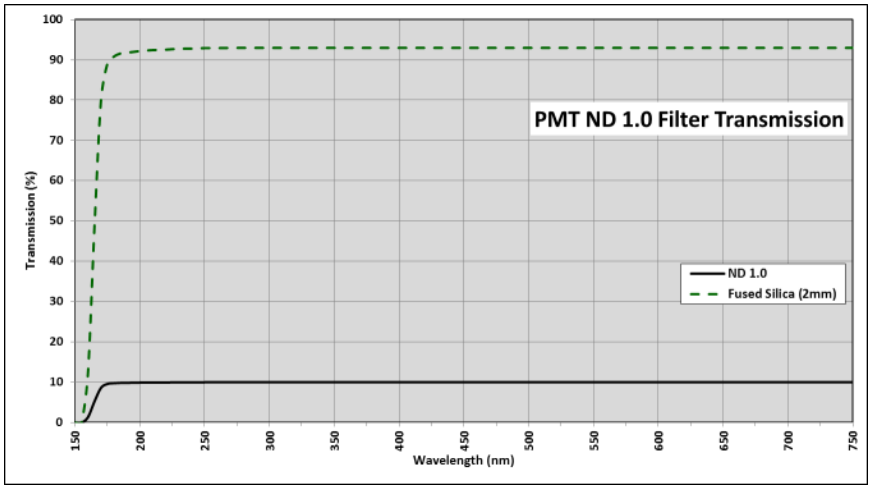


FIGURE 7. ND 1.0 Filter Curve

4.1.1.6 Filter Assembly 3 – CH* Filter

Filter Assembly 3 associated with PMT Module 3 contains a custom Narrow Bandpass Filter, provided by Alluxa, Inc. This filter has a Fused Silica substrate with a Narrow Bandpass coating. The coating center wavelength is ~ 432 nm with a FWHM (Full Width at Half Max) of 18 nm. See the filter curve and data below.

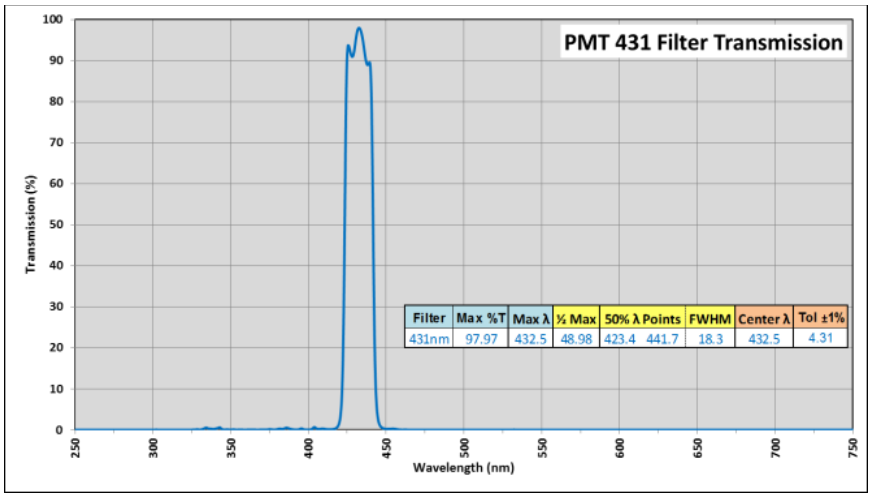


FIGURE 8. CH* Filter Curve

4.1.1.7 PMT Spectral Summary

A graphical summary of the spectral characteristics of the PMT Assembly is shown below.

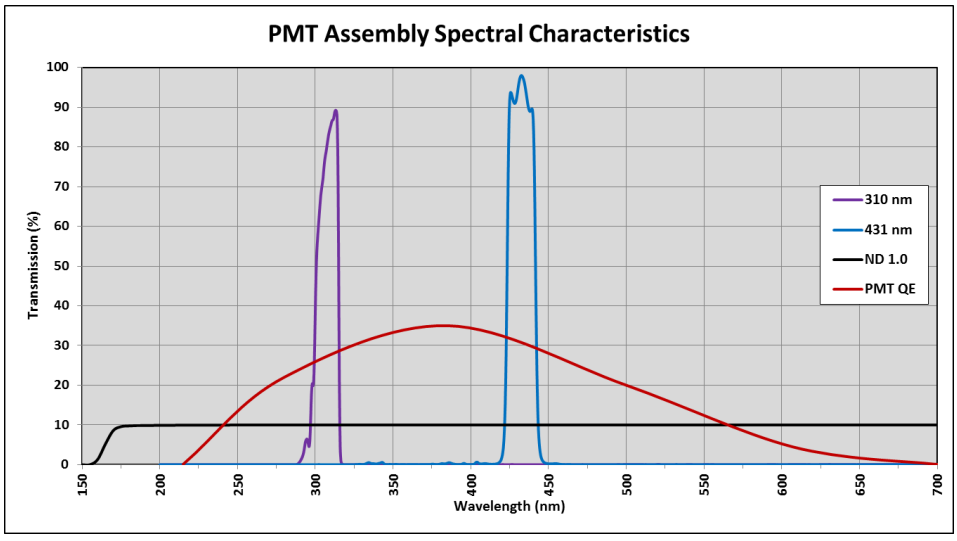


FIGURE 9. PMT Assembly Spectral Summary

4.1.1.8 Test Configuration

PMTs are extremely sensitive light detectors, providing an output proportional to incoming light intensity. Characterization is typically carried out using a calibrated tungsten or similar light source. ACME however uses the PMTs as a monitor for combustion events and in particular for flame extinguishment, thus a more suitable source for characterization would be temperature based such as the calibrated Blackbody temperature source that was used for the ACME PMT characterization.

The ACME PMT Assembly and the Blackbody source are positioned on an optical bench. The Blackbody source output was aligned, via optical mounting hardware, with the opening of the particular PMT location (1, 2 or 3) being tested.

The optical mounting hardware allows for alignment of the PMT Assembly in both the X and Y directions and also sets the distance from the outer surface of each PMT to the Blackbody aperture wheel (Z direction), equal to the appropriate flight distance of 112.5 mm.

All testing was done under blackout curtains that were arranged such that only the output from the Blackbody was visible during characterization. See the picture below for the full setup (Note: side curtains are raised in order to obtain the picture).

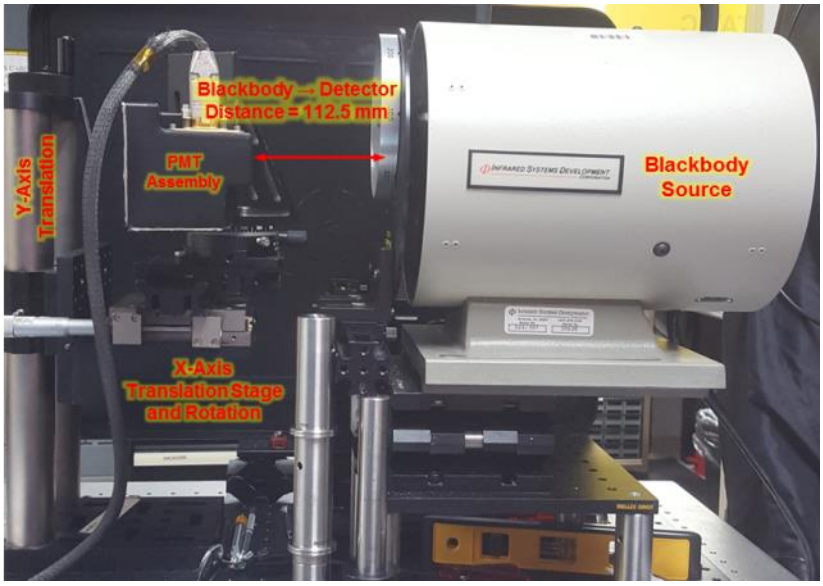


FIGURE 10. PMT Assembly Alignment Setup

To reduce the error caused by the Blackbody Aperture Wheel heating up over time and emitting unintended radiation, all Blackbody characterizations in this document were run with the Aperture Wheel mounted it on a slide rail, allowing it to be retracted away from the hot Blackbody source between measurements. By limiting the time resident near the hot Blackbody source, the aperture wheel could be held at room temperature.

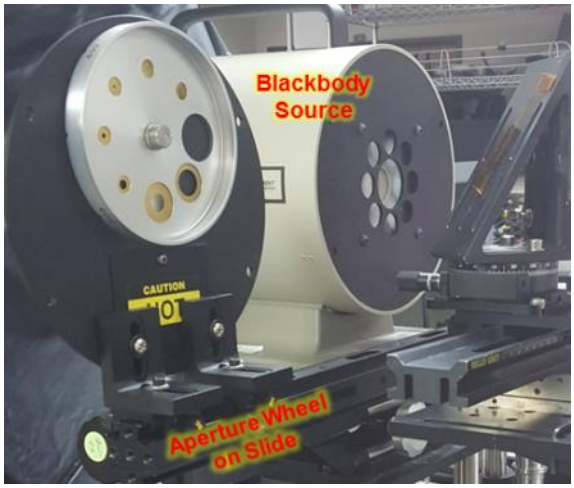


FIGURE 11. Aperture Wheel on Translation Slide

The PMT Assembly is connected for both power in and control via the Upstream Relay Board Test setup as shown in the figure below. Further details on the specific connections of this setup can be found in ACME-TST-031, "Upstream Relay Connector Board Test Procedure."

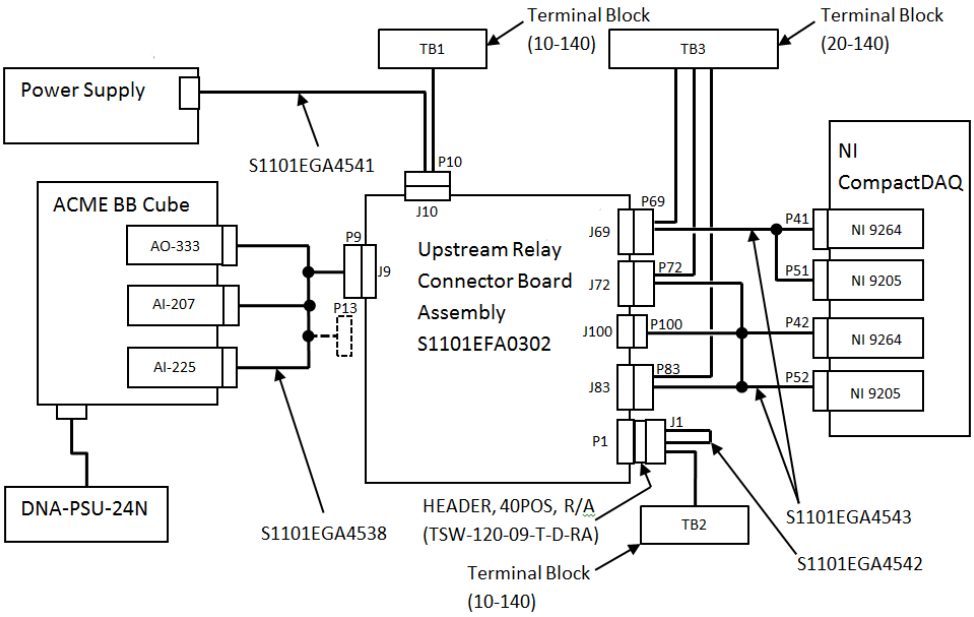


FIGURE 12. Cabling Schematic for Characterization Setup

A voltmeter was used to monitor the output signal (V_{out}) of each individual PMT. An additional voltmeter was connected to the Power Supply to ensure the input voltages remained within the required range. The DC Power Supply, Voltmeters, Oscilloscope and Blackbody Temperature Controller are located on a bench adjacent to the PMT and Blackbody (on the optical bench behind this shelf). Below is a photo of the support equipment in the laboratory setting.

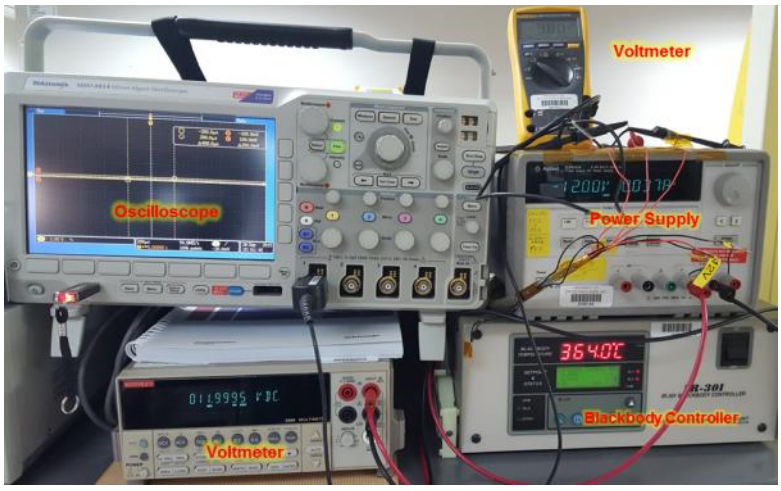


FIGURE 13. GSE for Characterization Setup

5.0 CHARACTERIZATION RESULTS

5.1 Dark Current

5.1.1 Dark Current Setup

Dark Current refers to the small amount of current resulting from a PMT even when operating in complete darkness. It is the lower limit to the current that can be measured by the PMT.

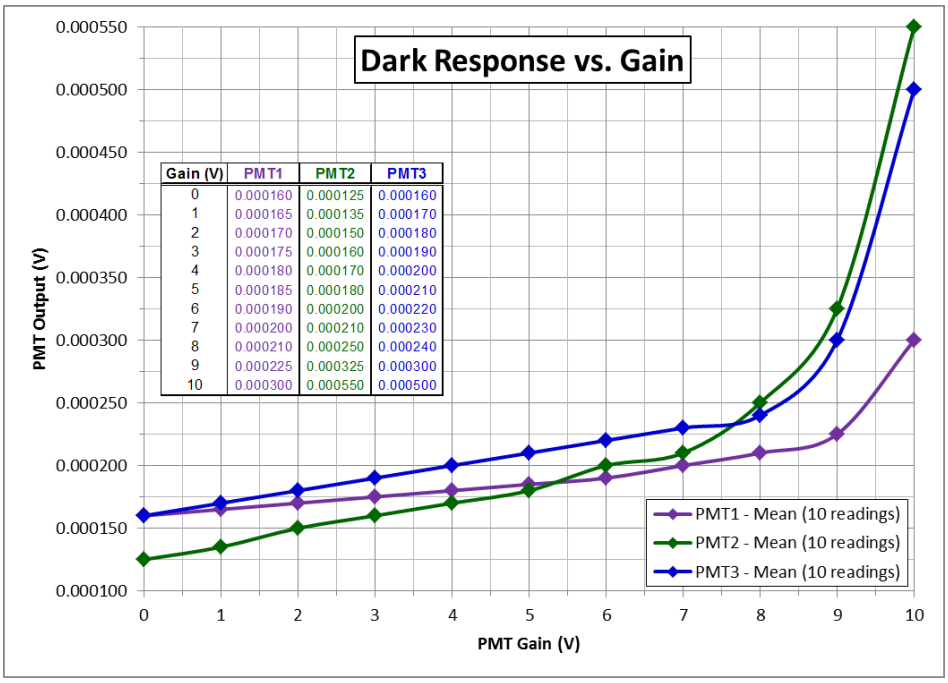
The Dark Current varies with the Control Voltage in proportion to the change in Gain. Thus the PMT Response over the range of Gain values was measured separately for each of the 3 PMT Modules. Each PMT was physically covered and shielded from light inside the dark enclosure.

The PMT Assembly was powered. With the PMT Gain set to 0 V, using a calibrated multi-meter, 10 consecutive Output (V) readings were taken and then averaged. In addition, both the High and Low Output (V) readings were recorded.

The PMT Gain was then increased to 1.0 V and the process was repeated. This continued in 1.0 V increments for PMT Gains up to the maximum of 10 V. This process was performed for each of the 3 PMTs.

5.1.2 Dark Response Data / Analysis

The resulting data and curves represent the mean of 10 consecutive Output Voltage readings at each Gain setting.



Gain (V)	PMT1	PMT2	PMT3
0	0.000160	0.000125	0.000160
1	0.000165	0.000135	0.000170
2	0.000170	0.000150	0.000180
3	0.000175	0.000160	0.000190
4	0.000180	0.000170	0.000200
5	0.000185	0.000180	0.000210
6	0.000190	0.000200	0.000220
7	0.000200	0.000210	0.000230
8	0.000210	0.000250	0.000240
9	0.000225	0.000325	0.000300
10	0.000300	0.000550	0.000500

FIGURE 14. PMT Mean Dark Response (10 consecutive pts) vs. Gain

Additionally a simple average of the maximum and minimum recorded Output Voltage readings was calculated for each Gain setting. These values were the same for all 3 PMT units and the resulting curve is shown along with the previous data in the figure below.

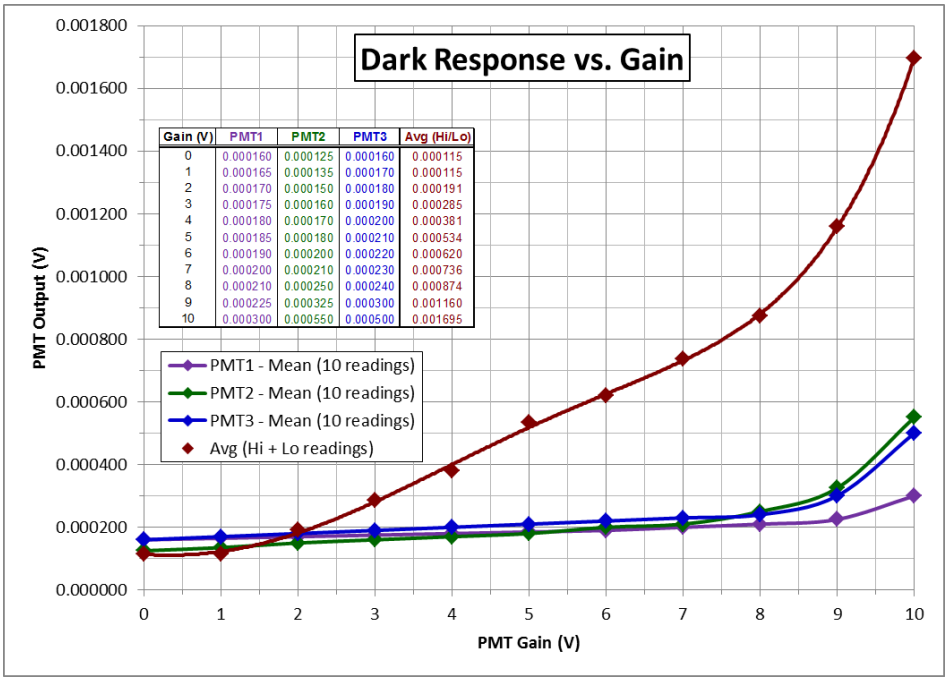


FIGURE 15. PMT Dark Response (Avg of Min & Max readings) vs. Gain

5.2 Field of View

5.2.1 FOV Setup

The Field of View (FOV) is performed separately for each of the 3 PMT Modules. Each PMT is centered and aligned normal to the Blackbody source, at a distance of 112.5 mm from the Blackbody Aperture to the outer PMT face. Each PMT is then translated in the horizontal (X) axis over a range of ±70 mm about the center position, in 10 mm increments, while recording the output signal level. This determines the X axis FOV and roll-off characteristics.

While at the X axis center point, the vertical (Y) axis characterization is measured by raising/lowering the PMT on a lab jack over ±70 mm to measure the Y axis FOV and roll-off characteristics.

5.2.2 PMT1 Field of View Data / Analysis

Prior to the measurements, PMT1 was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting PMT1 response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the PMT apertures) and Y (vertical, along the axis perpendicular to the line bisecting the PMT apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for PMT1.

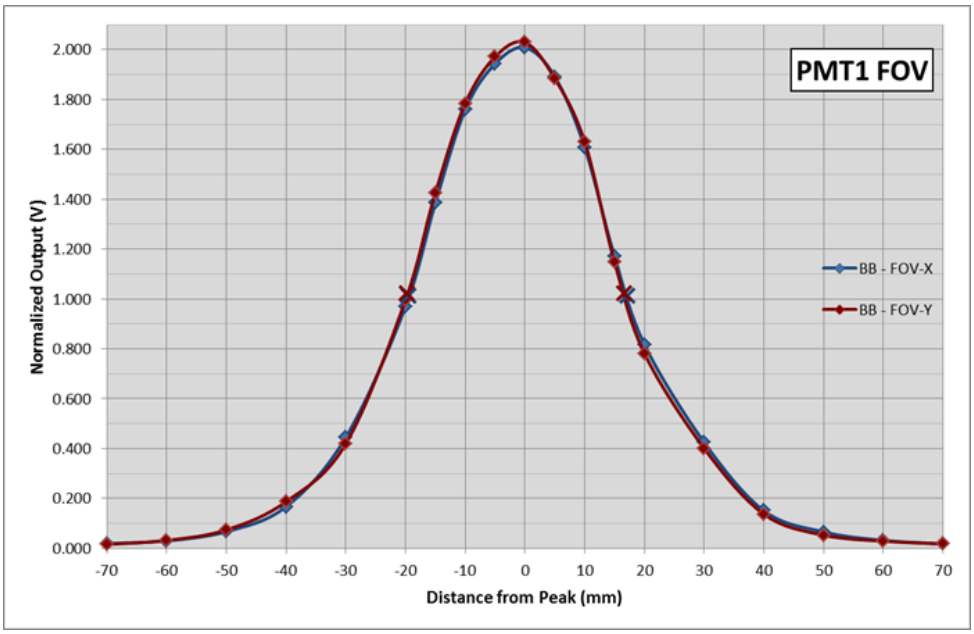


FIGURE 16. PMT1 (310nm) – FOV Characterization

The following parameter settings were used for the PMT1 FOV scans:

Blackbody Temperature = 1000°C
Blackbody Aperture = 0.4”
PMT1 Gain = 6.8 V

The following table shows the data resulting from the FOV scans for PMT1.

TABLE V. Data for PMT1 (310nm) FOV

PMT1 - X Stats:			PMT1 - Y Stats:		
Pk to Pk	1.99	V	Pk to Pk	2.01	V
V_{FWHM}	1.01	V	V_{FWHM}	1.02	V
X_1	-19.5	mm	Y_1	-19.7	mm
X_2	17.2	mm	Y_2	16.7	mm
FWHM	36.7	mm	FWHM	36.4	mm
$Center_{FWHM}$	**-1.1**	**mm**	**$Center_{FWHM}$**	**-1.5**	**mm**
$V_{90\%}$	1.81	V	$V_{90\%}$	1.83	V
$X_{1\text{-}90\%}$	-8.6	mm	$Y_{1\text{-}90\%}$	-8.8	mm
$X_{2\text{-}90\%}$	6.5	mm	$Y_{2\text{-}90\%}$	6.1	mm
$FW_{90\%}$	15.1	mm	$FW_{90\%}$	14.9	mm
$Center_{Alignment}$	**-1.1**	**mm**	**$Center_{Alignment}$**	**-1.4**	**mm**
V_{FOV}	0.038	V	V_{FOV}	0.036	V
$X_{1\text{-}FOV}$	-57.9	mm	$Y_{1\text{-}FOV}$	-59.0	mm
$X_{2\text{-}FOV}$	58.3	mm	$Y_{2\text{-}FOV}$	57.0	mm
FOV	**116.1**	**mm**	**FOV**	**116.0**	**mm**
$Center_{FOV}$	0.2	mm	$Center_{FOV}$	-1.0	mm

The results for PMT1 show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of -1.1 mm and -1.5 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of PMT1 to the measurement location:

PMT1 Alignment: X = -1.1 mm and Y = -1.4 mm

The PMT devices have shown extreme sensitivity and the ability to still detect usable signal at 99% roll off that is much greater than the measured dark response at the corresponding Gain (V). Thus the 1% points were used to determine the overall Field of View:

PMT1 FOV: X = 116.1 mm and Y = 116.0 mm

5.2.3 PMT 2 Field of View Data / Analysis

Prior to the measurements, PMT2 was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting PMT2 response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the PMT apertures) and Y (vertical, along the axis perpendicular to the line bisecting the PMT apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for PMT2.

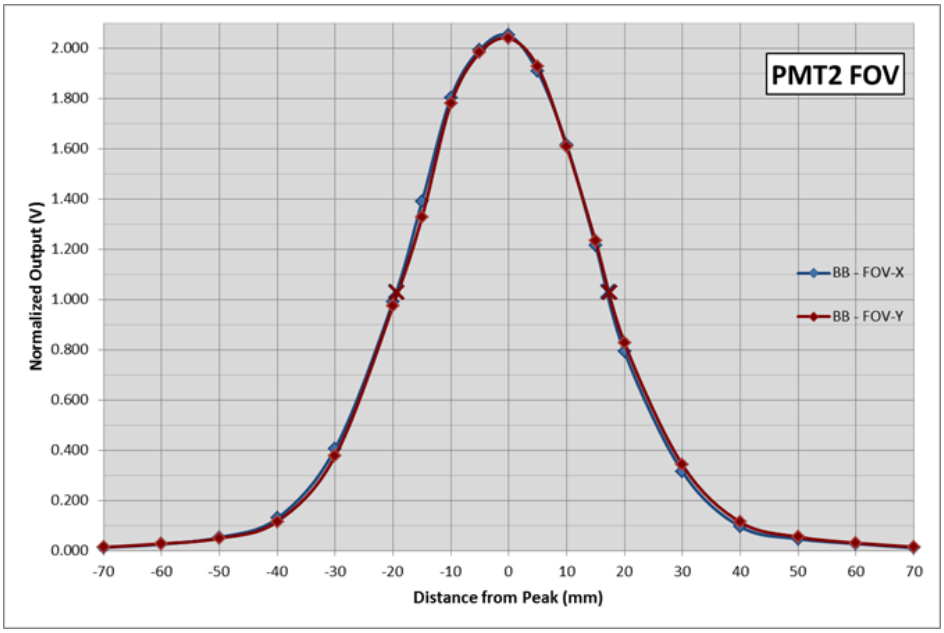


FIGURE 17. PMT2 (ND 1.0) – FOV Characterization

The following parameter settings were used for the PMT2 FOV scans:

Blackbody Temperature = 800°C
Blackbody Aperture = 0.4”
PMT2 Gain = 5.5 V

The following table shows the data resulting from the FOV scans for PMT2.

TABLE VI. Data for PMT 2 (ND 1.0) FOV

PMT2 - X Stats:			PMT2 - Y Stats:		
Pk to Pk	2.04	V	Pk to Pk	2.03	V
V_{FWHM}	1.03	V	V_{FWHM}	1.03	V
X_1	-19.5	mm	Y_1	-19.3	mm
X_2	17.2	mm	Y_2	17.5	mm
FWHM	36.7	mm	FWHM	36.8	mm
$Center_{FWHM}$	**-1.2**	**mm**	**$Center_{FWHM}$**	**-0.9**	**mm**
$V_{90\%}$	1.85	V	$V_{90\%}$	1.84	V
$X_{1\text{-}90\%}$	-8.8	mm	$Y_{1\text{-}90\%}$	-8.6	mm
$X_{2\text{-}90\%}$	6.0	mm	$Y_{2\text{-}90\%}$	6.4	mm
$FW_{90\%}$	14.9	mm	$FW_{90\%}$	15.0	mm
$Center_{Alignment}$	**-1.4**	**mm**	**$Center_{Alignment}$**	**-1.1**	**mm**
V_{FOV}	0.031	V	V_{FOV}	0.034	V
$X_{1\text{-}FOV}$	-58.0	mm	$Y_{1\text{-}FOV}$	-57.0	mm
$X_{2\text{-}FOV}$	58.2	mm	$Y_{2\text{-}FOV}$	58.6	mm
FOV	**116.2**	**mm**	**FOV**	**115.7**	**mm**
$Center_{FOV}$	0.1	mm	$Center_{FOV}$	0.8	mm

The results for PMT2 show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of -1.2 mm and -0.9 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of PMT2 to the measurement location:

PMT2 Alignment: X = -1.4 mm and Y = -1.1 mm

The PMT devices have shown extreme sensitivity and the ability to still detect usable signal at 99% roll off that is much greater than the measured dark response at the corresponding Gain (V). Thus the 1% points were used to determine the overall Field of View:

PMT2 FOV: X = 116.2 mm and Y = 115.7 mm

5.2.4 PMT 3 Field of View Data / Analysis

Prior to the measurements, PMT3 was aligned to the Blackbody source, by locating the peak of the signal output at the current Gain setting. The resulting PMT3 response to off-axis translations is shown in the figure below. It shows the X (horizontal, along the axis parallel to the line bisecting the PMT apertures) and Y (vertical, along the axis perpendicular to the line bisecting the PMT apertures) FOV and roll-off measurements.

The following curves show the results from the FOV scans in both the X and Y directions for PMT3.

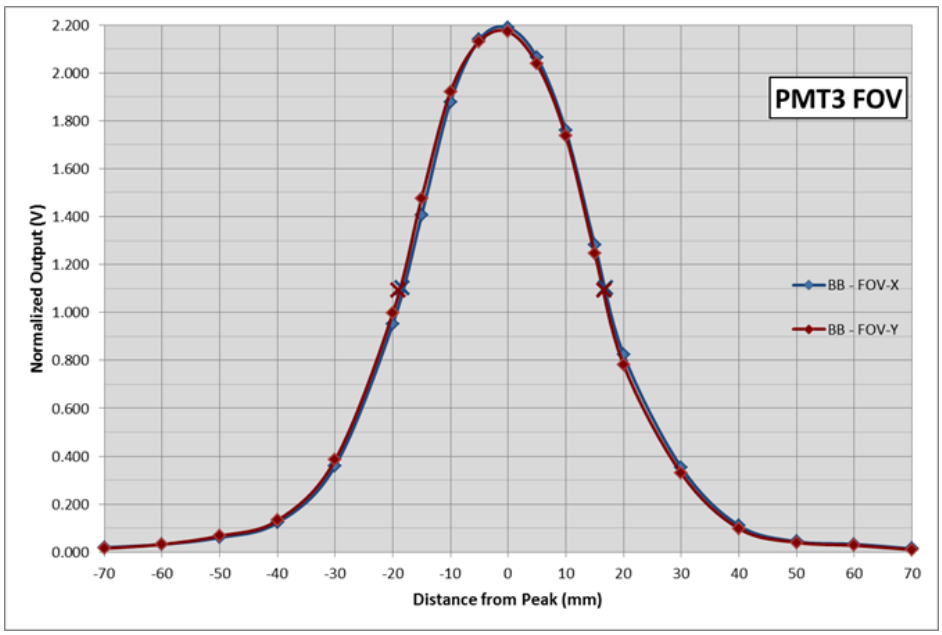


FIGURE 18. PMT3 (431nm) – FOV Characterization

The following parameter settings were used for the PMT2 FOV scans:

Blackbody Temperature = 1000°C
Blackbody Aperture = 0.4”
PMT2 Gain = 5.9 V

The following table shows the data resulting from the FOV scans for PMT3.

TABLE VII. Data for PMT 3 (431nm) FOV

PMT3 - X Stats:			PMT3 - Y Stats:		
Pk to Pk	2.18	V	Pk to Pk	2.16	V
V_{FWHM}	1.10	V	V_{FWHM}	1.09	V
X_1	-18.4	mm	Y_1	-19.0	mm
X_2	17.0	mm	Y_2	16.7	mm
FWHM	35.3	mm	FWHM	35.7	mm
$Center_{FWHM}$	**-0.7**	**mm**	**$Center_{FWHM}$**	**-1.2**	**mm**
$V_{90\%}$	1.97	V	$V_{90\%}$	1.96	V
$X_{1\text{-}90\%}$	-8.2	mm	$Y_{1\text{-}90\%}$	-9.2	mm
$X_{2\text{-}90\%}$	6.5	mm	$Y_{2\text{-}90\%}$	6.4	mm
$FW_{90\%}$	14.7	mm	$FW_{90\%}$	15.5	mm
$Center_{Alignment}$	**-0.9**	**mm**	**$Center_{Alignment}$**	**-1.4**	**mm**
V_{FOV}	0.036	V	V_{FOV}	0.031	V
$X_{1\text{-}FOV}$	-58.7	mm	$Y_{1\text{-}FOV}$	-60.2	mm
$X_{2\text{-}FOV}$	57.5	mm	$Y_{2\text{-}FOV}$	56.7	mm
FOV	**116.2**	**mm**	**FOV**	**116.9**	**mm**
$Center_{FOV}$	-0.6	mm	$Center_{FOV}$	-1.8	mm

The results for PMT3 show that the center of symmetry does not completely agree with the geometric peak position, as indicated by the X_{center} and Y_{center} fits of -0.7 mm and -1.2 mm, respectively. These centers were calculated by determining the Full Width at Half Maximum (FWHM) points on each curve and interpolating to get the position of each FWHM point.

The 90% points (10% roll-off) were used to verify the alignment of PMT3 to the measurement location:

PMT3 Alignment: X = -0.9 mm and Y = -1.4 mm

The PMT devices have shown extreme sensitivity and the ability to still detect usable signal at 99% roll off that is much greater than the measured dark response at the corresponding Gain (V). Thus the 1% points were used to determine the overall Field of View:

PMT3 FOV: X = 116.2 mm and Y = 116.9 mm

5.3 Radiometric Calibration

5.3.1 Radiometric Calibration Setup

The Radiometric Calibration is performed separately for the 3 PMTs at various Input Gain values. Each PMT is centered and aligned normal to the Blackbody source, at the Flight distance of 112.5 mm from the Blackbody Aperture to the outer PMT face.

Within each of the Radiometer Calibrations, the PMT Input Gain is set to a discrete value. Then the PMT Output Voltage was recorded over a range of Blackbody Apertures for each Blackbody Temperature. This simulates various source Irradiance levels covering the expected operating range.

The Table below shows the expected Irradiance Range matrix for the calibration of the PMTs.

TABLE VIII. Expected Range for Radiometric Calibration

Temp. (°C)	Aperture Diameter					
	0.6	0.4	0.2	0.1	0.05	0..025
700's		10% 1.27E-02		0.5% 6.48E-04		Min 4.05E-05
800's	33% 4.14E-02			1% 1.25E-03		
900's	50% 6.21E-02			5% 6.26E-03		
1000's	66% 8.21E-02					
1100's		33% 4.11E-02	10% 1.25E-02		0.5% 6.42E-04	
1200	Max 1.22E-01					

5.3.2 PMT1 Radiometric Calibration Data / Analysis

Using the setup described above, the Radiometric Calibration data, at various Input Gain settings for PMT1, were collected and shown in the Table below. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE IX. PMT1 Radiometric Calibration Data

Temp. (°C)	Apert (inch)	Irradiance (W/cm^2)	PMT 1 (310nm) Output									
			2V	3V	4V	5V	5.5V	6V	7V	8V	9V	10V
800	0.025	**0.0000598**				0.000450	0.000875	0.001499	0.005519	0.017517	0.046644	0.104801
900	0.025	0.0000855				0.000515	0.001035	0.001954	0.007287	0.023507	0.063224	0.142153
1000	0.025	0.0001186			0.000150	0.000526	0.001151	0.002276	0.008373	0.033696	0.063572	0.144172
1100	0.025	0.0001604			0.000223	0.000773	0.001549	0.003325	0.019738	0.041555	0.110159	0.242741
1200	0.025	0.0002125			0.000276	0.000861	0.001838	0.003814	0.025286	0.058913	0.130301	0.288082
800	0.05	0.0002394			0.000281	0.000947	0.002037	0.004126	0.026294	0.062328	0.139589	0.309508
900	0.05	0.0003419			0.000407	0.001874	0.003546	0.006218	0.036937	0.119756	0.319553	0.706438
1000	0.05	0.0004742			0.000468	0.002431	0.005682	0.009932	0.047645	0.135771	0.414790	0.817829
1100	0.05	0.0006417			0.000476	0.002527	0.006806	0.012085	0.058278	0.157502	0.416133	0.918512
1200	0.05	0.0008501			0.000494	0.003812	0.010695	0.018578	0.070336	0.264013	0.436494	0.962811
800	0.1	0.0009576		0.000150	0.000962	0.004650	0.012333	0.020015	0.121551	0.397620	1.064071	2.346491
900	0.1	**0.0013675**		0.000249	0.001086	0.006296	0.016235	0.034174	0.136947	0.449194	1.186206	2.608219
1000	0.1	0.0018969		0.000261	0.001302	0.008439	0.019998	0.045556	0.172412	0.512981	1.396997	3.254296
1100	0.1	0.0025668		0.000283	0.001331	0.009457	0.028564	0.063670	0.177656	0.581312	1.538948	3.333056
1200	0.1	0.0034002		0.000305	0.001496	0.013291	0.035839	0.078386	0.196523	0.637304	1.793753	3.632010
800	0.2	0.0038302		0.000398	0.001548	0.014790	0.039123	0.103455	0.218792	0.711875	2.002978	3.978887
900	0.2	0.0054701		0.000463	0.003581	0.020977	0.055497	0.130017	0.326184	1.410404	4.013261	4.600000
1000	0.2	0.0075875		0.000508	0.004136	0.025169	0.068938	0.152707	0.421712	1.824833	4.555716	
1100	0.2	0.0102673		0.000558	0.004684	0.032553	0.091510	0.170466	0.689142	2.241431	4.600000	
1200	0.2	**0.0136010**		0.000639	0.006126	0.045762	0.111165	0.206567	0.858276	2.974719		
800	0.4	0.0153209	0.000150	0.000822	0.007412	0.052429	0.154011	0.275593	1.124935	3.665706		
900	0.4	0.0218805	0.000218	0.001414	0.014361	0.102319	0.219063	0.435807	1.673729	4.600000		
1000	0.4	0.0303501	0.000224	0.001702	0.017449	0.124588	0.389950	0.695312	2.659576			
800	0.6	0.0344720	0.000237	0.001757	0.018267	0.129608	0.494003	0.878398	3.539043			
1100	0.4	0.0410690	0.000271	0.002883	0.031272	0.224557	0.701302	1.178047	4.600000			
900	0.6	0.0492311	0.000332	0.004987	0.055351	0.397132	1.040605	1.884077				
1200	0.4	0.0544039	0.000362	0.005872	0.065149	0.467564	1.254513	2.441461				
1000	0.6	**0.0682878**	0.000468	0.010265	0.116087	0.738510	2.159842	4.407241				
1100	0.6	0.0924053	0.000674	0.016737	0.189516	1.365642	4.096154	4.600000				
1200	0.6	**0.1224089**	0.001082	0.029368	0.433631	2.894306	4.600000					

The PMT1 data was graphed (shown in the Figure below), independent of the aperture/temperature combination in order to produce the corresponding Irradiance curves. The results show the correlation between Output Voltage and Irradiance for each Gain input to PMT1.

It is clear upon viewing the graphical representation of the PMT1 data at the various Gain settings, that polynomial curve fits are appropriate. The overlay of data fit by method of least squares is shown for each curve.

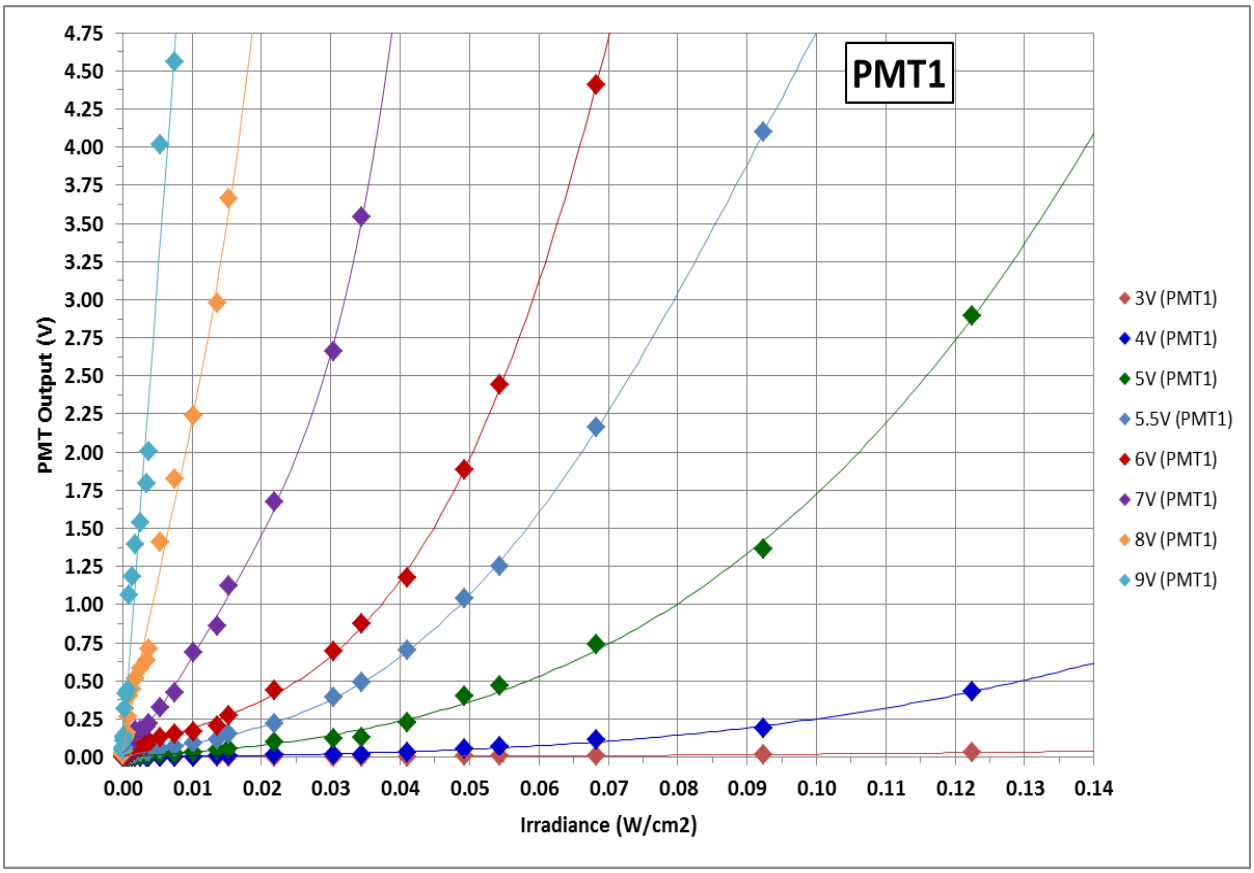


FIGURE 19. PMT1 (310nm) Characterization Curves

The equations corresponding to the curve fits from the above PMT1 data are shown in the Table below:

TABLE X. PMT1 Curve Fit Equations

PMT1 (310nm)	
Gain = 3V	$y = 1.991881x^2 - 0.005211x + 0.000200$
Gain = 4V	$y = 196.835980x^3 + 0.345545x^2 + 0.519914x + 0.000150$
Gain = 5V	$y = 1{,}104.055224x^3 + 34.407322x^2 + 2.770232x + 0.000277$
Gain = 5.5V	$y = -58{,}849.143398x^4 + 11{,}868.553546x^3 - 224.738423x^2 + 10.187086x + 0.000282$
Gain = 6V	$y = -55{,}617.012418x^4 + 22{,}186.536267x^3 - 645.571330x^2 + 22.895739x + 0.000400$
Gain = 7V	$y = 3{,}308{,}609.058167x^4 - 166{,}821.231989x^3 + 3{,}599.519526x^2 + 39.920446x + 0.003750$
Gain = 8V	$y = 395{,}393.992726x^3 - 7{,}363.493027x^2 + 253.097826x + 0.002500$

5.3.3 PMT2 Radiometric Calibration Data / Analysis

Using the setup described above, the Radiometric Calibration data, at various Input Gain settings for PMT2, were collected and shown in the Table below. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XI. PMT2 Radiometric Calibration Data

Temp. (°C)	Apert (inch)	Irradiance (W/cm^2)	PMT 2 (ND 1.0) Output										
			1V	1.5V	2V	2.5V	3V	3.5V	4V	5V	6V	7V	8V
700	0.025	0.0000405						0.000361	0.000721	0.004233	0.021604	0.085118	0.271433
800	0.025	0.0000598					0.000120	0.000541	0.000962	0.005661	0.028534	0.112689	0.365801
900	0.025	0.0000855				0.000120	0.000267	0.000827	0.001387	0.008947	0.045263	0.178527	0.566191
1000	0.025	0.0001186				0.000140	0.000279	0.000984	0.001688	0.010878	0.056311	0.222352	0.698059
1100	0.025	0.0001604				0.000186	0.000371	0.001205	0.002839	0.019357	0.098758	0.393342	1.248519
700	0.05	0.0001619				0.000277	0.000554	0.001740	0.004525	0.031388	0.161637	0.644473	2.050252
1200	0.025	0.0002125				0.000307	0.000614	0.002276	0.005337	0.037053	0.230262	0.754947	2.391653
800	0.05	0.0002394				0.000494	0.000987	0.003292	0.009597	0.066926	0.343747	1.369221	4.306523
900	0.05	0.0003419			0.000120	0.000768	0.001415	0.004902	0.014389	0.101384	0.523776	2.091912	4.590000
1000	0.05	0.0004742			0.000183	0.000826	0.001469	0.006296	0.015122	0.106071	0.644658	2.267773	
1100	0.05	0.0006417			0.000188	0.000905	0.001622	0.008288	0.016953	0.119458	0.714627	2.442301	
700	0.1	0.0006475			0.000192	0.001277	0.002361	0.011827	0.025292	0.178754	0.922306	3.674056	
1200	0.05	0.0008501			0.000229	0.001729	0.003228	0.015233	0.035237	0.249462	1.288287	4.590000	
800	0.1	0.0009576			0.000299	0.002731	0.005163	0.020108	0.047053	0.403299	1.879404		
900	0.1	0.0013675			0.000303	0.002966	0.005629	0.024007	0.062385	0.441344	2.273467		
1000	0.1	0.0018969			0.000320	0.003000	0.005679	0.036354	0.073029	0.455952	2.997019		
1100	0.1	0.0025668		0.000120	0.000486	0.005777	0.009068	0.053939	0.124810	0.785314	4.547863		
700	0.2	0.0025900		0.000196	0.000701	0.005807	0.010913	0.058213	0.125512	1.018437	4.590000		
1200	0.1	0.0034002		0.000202	0.000708	0.007026	0.012943	0.085062	0.193181	1.373105			
800	0.2	0.0038302		0.000205	0.000808	0.007868	0.013127	0.094730	0.216332	1.597048			
900	0.2	0.0054701		0.000222	0.000812	0.011068	0.019323	0.144559	0.289795	1.789694			
1000	0.2	0.0075875		0.000256	0.000831	0.014793	0.025754	0.200992	0.456230	2.918436			
1100	0.2	0.0102673		0.000346	0.002103	0.028358	0.046612	0.339258	0.631904	4.569633			
700	0.4	0.0103600		0.000374	0.002262	0.029640	0.049017	0.363641	0.678264	4.590000			
1200	0.2	0.0136010		0.000422	0.002789	0.047887	0.076984	0.440723	0.884462				
800	0.4	0.0153209		0.000437	0.002819	0.059967	0.087115	0.536546	0.986217				
900	0.4	0.0218805	0.000120	0.000508	0.003641	0.082949	0.164257	0.849567	1.794877				
700	0.6	0.0233101	0.000188	0.001048	0.008749	0.134308	0.239866	0.913486	2.273748				
1000	0.4	0.0303501	0.000191	0.001055	0.009172	0.157792	0.306412	1.414663	3.291975				
800	0.6	0.0344720	0.000196	0.001173	0.010292	0.201381	0.432468	1.794654	3.783519				
1100	0.4	0.0410690	0.000235	0.001742	0.015769	0.336704	0.617639	2.280654	4.590000				
900	0.6	0.0492311	0.000317	0.003294	0.031461	0.538202	1.004942	2.967471					
1200	0.4	0.0544039	0.000372	0.003467	0.032951	0.664603	1.096254	3.865184					
1000	0.6	0.0682878	0.000488	0.005515	0.054492	0.949502	1.844512	4.590000					
1100	0.6	0.0924053	0.000644	0.009333	0.093682	1.459692	3.325701						
1200	0.6	0.1224089	0.000894	0.013019	0.143038	2.696519	4.590000						

The PMT2 data was graphed (shown in the Figure below), independent of the aperture/temperature combination in order to produce the corresponding Irradiance curves. The results show the correlation between Output Voltage and Irradiance for each Gain input to PMT2.

It is clear upon viewing the graphical representation of the PMT2 data at the various Gain settings, that polynomial curve fits are appropriate. The overlay of data fit by method of least squares is shown for each curve.

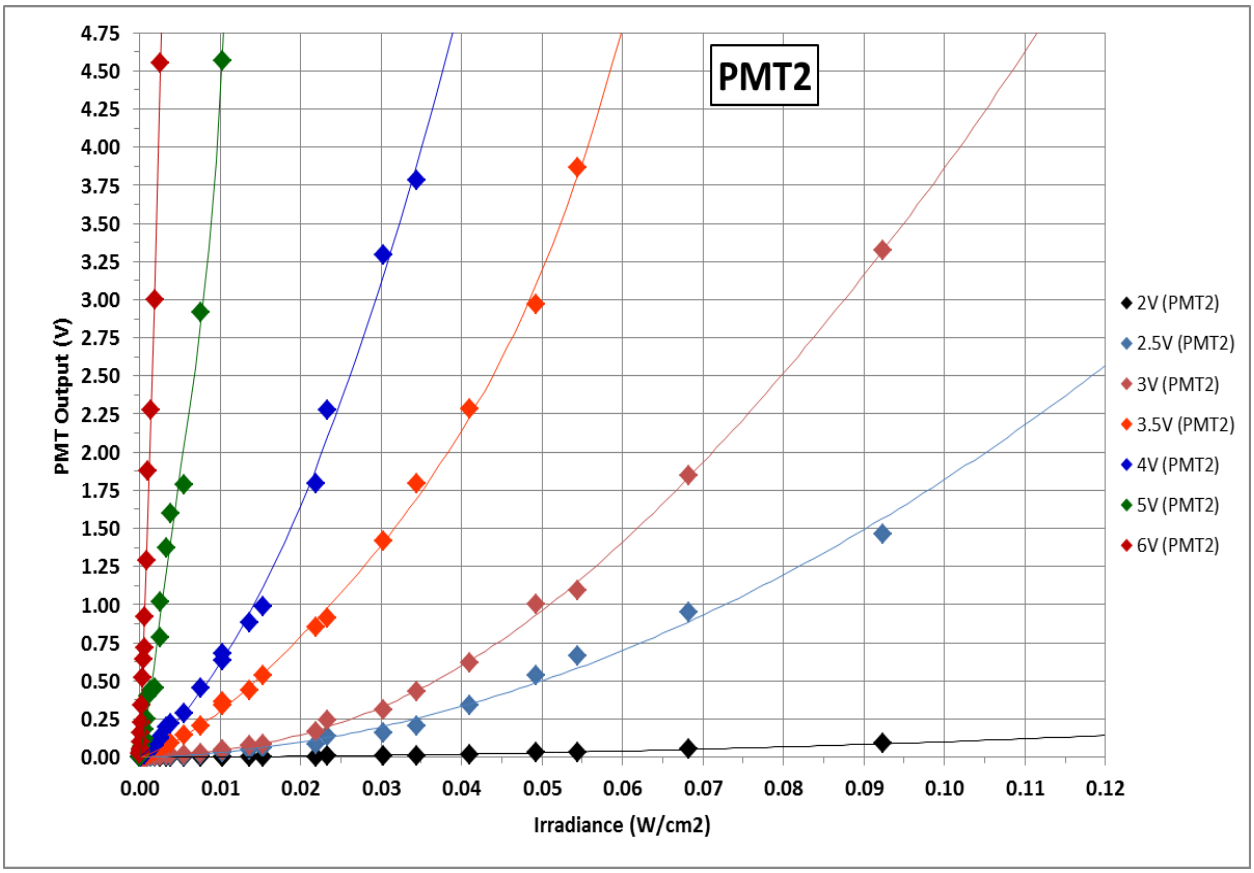


FIGURE 20. PMT2 (ND 1.0) Characterization Curves

The equations corresponding to the curve fits from the above PMT2 data are shown in the Table below:

TABLE XII. PMT2 Curve Fit Equations

PMT2 (ND 1.0)	
Gain = 2V	$y = 8.585779x^2 + 0.156690x + 0.000105$
Gain = 2.5V	$y = -43.824964x^3 + 169.992980x^2 + 1.665894x - 0.000060$
Gain = 3V	$y = 291{,}761.896381x^5 - 88{,}314.764348x^4 + 8{,}801.002262x^3 + 65.786376x^2 + 3.172557x + 0.000010$
Gain = 3.5V	$y = 459{,}377.096347x^4 - 39{,}905.504054x^3 + 1{,}814.542253x^2 + 15.361730x + 0.000260$
Gain = 4V	$y = 2{,}092.641446x^2 + 40.832842x - 0.000930$
Gain = 5V	$y = 728{,}809{,}327.944092x^4 - 11{,}745{,}484.619257x^3 + 60{,}902.178128x^2 + 271.865031x - 0.007000$
Gain = 6V	$y = 71{,}216.045377x^2 + 1{,}600.544829x - 0.043000$

5.3.4 PMT3 Radiometric Calibration Data / Analysis

Using the setup described above, the Radiometric Calibration data, at various Input Gain settings for PMT3, were collected and shown in the Table below. The grayed out boxes indicate the readings were out of range, either on the low or high end, and have been excluded from the measurements and plots.

TABLE XIII. PMT3 Radiometric Calibration Data

Temp. (°C)	Apert (inch)	Irradiance (W/cm^2)	PMT 3 (431nm) Output									
			2V	3V	4V	4.5V	5V	6V	7V	8V	9V	10V
800	0.025	**0.0000598**			0.000724	0.002449	0.004174	0.022464	0.094367	0.308615	0.831654	1.803173
900	0.025	0.0000855			0.000856	0.002972	0.005087	0.025161	0.100213	0.326272	0.865370	1.934416
1000	0.025	0.0001186			0.000882	0.003073	0.005263	0.026017	0.107817	0.347521	0.927921	2.058096
1100	0.025	0.0001604			0.000894	0.003088	0.005282	0.026733	0.110869	0.351473	0.940127	2.140307
1200	0.025	0.0002125			0.001006	0.003939	0.006871	0.031371	0.126545	0.406918	1.082248	2.389925
800	0.05	0.0002394		0.000150	0.001108	0.004066	0.007023	0.035621	0.142580	0.445637	1.225592	2.661951
900	0.05	0.0003419		0.000216	0.001209	0.004524	0.007839	0.040241	0.161616	0.521964	1.389837	3.043429
1000	0.05	0.0004742		0.000228	0.001221	0.004542	0.007862	0.040919	0.162301	0.538741	1.395671	3.051593
1100	0.05	0.0006417		0.000239	0.001392	0.005142	0.008892	0.045734	0.182605	0.587075	1.561392	3.399806
1200	0.05	0.0008501		0.000283	0.001756	0.006689	0.011622	0.059866	0.240442	0.782828	2.061035	4.135699
800	0.1	0.0009576		0.000306	0.001914	0.007344	0.012774	0.065863	0.264485	0.854232	2.264566	4.288917
900	0.1	**0.0013675**		0.000409	0.002926	0.011559	0.020191	0.103751	0.417811	1.349609	3.567133	4.596000
1000	0.1	0.0018969		0.000414	0.002936	0.016622	0.030308	0.104542	0.419778	1.358224	3.581855	
1100	0.1	0.0025668		0.000657	0.005784	0.023174	0.040564	0.210462	0.843151	2.726457	4.596000	
1200	0.1	0.0034002		0.000912	0.008615	0.030618	0.050621	0.314648	1.258788	4.046584		
800	0.2	0.0038302		0.000916	0.008924	0.034055	0.058185	0.326285	1.311137	4.204166		
900	0.2	0.0054701		0.001129	0.013148	0.045227	0.079306	0.411576	1.648990	4.596000		
1000	0.2	0.0075875		0.001875	0.018396	0.069805	0.101214	0.524026	2.098849			
1100	0.2	0.0102673	0.000150	0.002443	0.024572	0.095738	0.146903	0.763376	3.067067			
1200	0.2	**0.0136010**	0.000252	0.003102	0.033868	0.138124	0.242379	1.159970	4.596000			
800	0.4	0.0153209	0.000259	0.003663	0.040644	0.166115	0.291585	1.516413				
900	0.4	0.0218805	0.000331	0.006054	0.068174	0.278740	0.489305	2.544181				
1000	0.4	0.0303501	0.000465	0.010102	0.114418	0.467512	0.820606	4.256556				
800	0.6	0.0344720	0.000496	0.011416	0.129972	0.631284	0.982595	4.596000				
1100	0.4	0.0410690	0.000598	0.014508	0.265608	0.976968	1.788327					
900	0.6	0.0492311	0.001289	0.037774	0.434065	1.476414	3.118762					
1200	0.4	0.0544039	0.001777	0.054991	0.564429	2.061536	4.558643					
1000	0.6	**0.0682878**	0.001977	0.062419	0.917491	3.991654	4.596000					
1100	0.6	0.0924053	0.005491	0.181243	2.093773	4.596000						
1200	0.6	**0.1224089**	0.009076	0.352136	4.086694							

The PMT3 data was graphed (shown in the Figure below), independent of the aperture/temperature combination in order to produce the corresponding Irradiance curves. The results show the correlation between Output Voltage and Irradiance for each Gain input to PMT3.

It is clear upon viewing the graphical representation of the PMT3 data at the various Gain settings, that polynomial curve fits are appropriate. The overlay of data fit by method of least squares is shown for each curve.

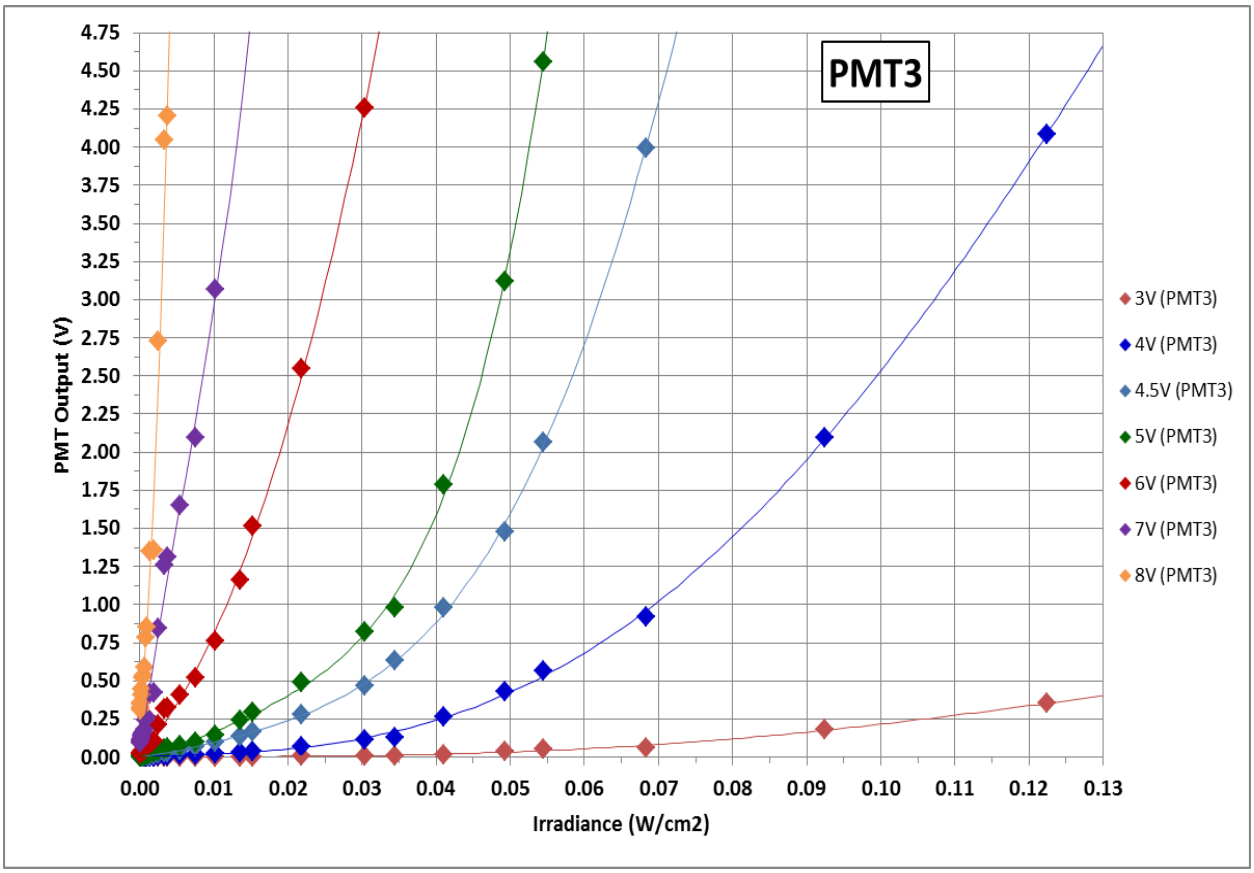


FIGURE 21. PMT3 (431nm) Characterization Curves

The equations corresponding to the curve fits from the above PMT3 data are shown in the Table below:

TABLE XIV. PMT3 Curve Fit Equations

PMT3 (431nm)	
Gain = 3V	$y = -1{,}764.688010x^4 + 508.863505x^3 - 15.508597x^2 + 0.393871x + 0.000070$
Gain = 4V	$y = -14{,}425.292616x^4 + 4{,}236.185250x^3 - 46.099632x^2 + 2.012753x + 0.000600$
Gain = 4.5V	$y = -2{,}955{,}659.559357x^5 + 543{,}397.470716x^4 - 19{,}411.363291x^3 + 505.687643x^2 + 5.655574x + 0.002100$
Gain = 5V	$y = 911{,}016.817407x^4 - 44{,}193.276271x^3 + 1{,}087.860546x^2 + 8.544647x + 0.003600$
Gain = 6V	$y = 7{,}785.072945x^3 + 2{,}717.730343x^2 + 50.108622x + 0.019400$
Gain = 7V	$y = 703{,}610.180561x^3 - 11{,}656.854323x^2 + 337.786210x + 0.074000$
Gain = 8V	$y = 165{,}632.254213x^2 + 451.923332x + 0.285000$

5.4 Response Time

5.4.1 Response Time Analysis

The PMT is a detector that has an exceptionally fast time response, on the order of nanoseconds (ns) or less. PMT time response is determined primarily by the Electron Transit Time (t_t), which is the time interval between the arrival of a light pulse at the photocathode and the instant when the anode output pulse reaches peak amplitude. The anode pulse Rise Time (t_r) is defined as the time required to rise from 10% to 90% of the peak amplitude. The Transit Time Spread (TTS) is the fluctuation between individual light pulses in the transit time and is also defined as the Full Width Half Max (FWHM) of this distribution. The graph below illustrates the Time Response characteristics of PMT signals.

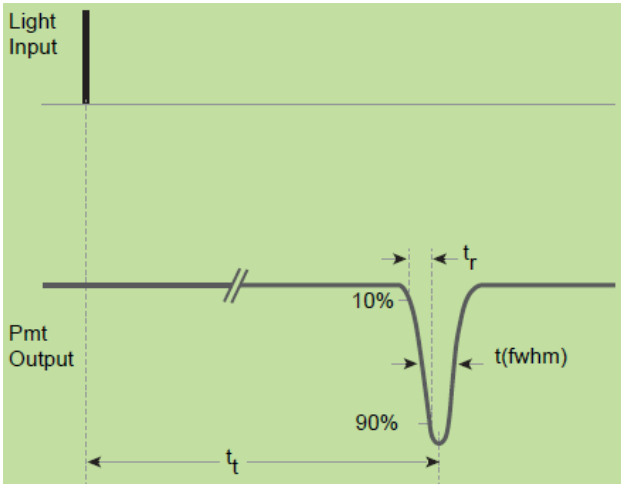


FIGURE 22. PMT Time Response

The Rise Time of a detector can typically be measured using a constant source with a method of pulsing the signal from the source, such as a chopper wheel. But for rise times faster than 10^{-5} seconds (< 0.1 ms) the pulses produced by choppers are not ideal shapes and unsatisfactory. Therefore for fast response detectors, such as PMTs, pulsed laser diodes with extremely short pulse widths are often used. Additionally, most standard oscilloscopes do not measure into the pico-second range necessary for rise times of PMTs.

For these reasons, the ACME PMT Response Time could not be measured directly. It is assumed that the response time is within the realm of vendor specifications shown in the Table below.

TABLE XV. ACME PMT Response Time Values

R9880U-110 time Response		
Rise Time	Transit Time	FWHM
0.57 ns	2.7 ns	0.2 ns

5.5 Recommended Baseline Settings

Each PMT has an adjustable Gain setting with a range of 0 – 10 VDC via the ACME Relay Board as described in Section 4.1.1.2.

The baseline Gain setting, based on Relay Board Input Voltage, for each PMT was determined in order to optimize for the widest overall Irradiance range. The recommended baseline Gain settings for each PMT are listed in the Table below.

TABLE XVI. ACME PMT Baseline Gain Values

Baseline Gain Setting		
PMT1	PMT2	PMT3
5.0 V	3.0 V	4.0 V

These settings are just a baseline. Adjusting the Gain appropriately allows the PMT to accommodate both strong and weak flames. For example, some test points will likely produce input intensity that is high enough to saturate a PMT at its optimal Gain setting. To get an optimum output from a particular PMT for these cases, it will be necessary to reduce the input Gain, even though it may result in decreased usable range on the top end.

APPENDIX A ACRONYMS AND ABBREVIATIONS

ACME	Advanced Combustion via Microgravity Experiments
BB	Broadband
CH*	Carbon-Hydrogen Radical
CIR	Combustion Integrated Rack
COTS	Commercial off The Shelf
DC	Direct Current
ERD	Engineering Requirements Document
ESD	Electrostatic Discharge
FCF	Fluids and Combustion Facility
FWHM	Full Width at Half Max
FOV	Field of View
FS	Fused Silica
GL	Gray Level
IR	Infrared
ISS	International Space Station
ms	Milliseconds
N/A	Not Applicable
NASA	National Aeronautics and Space Administration
NB	Narrowband
ND	Neutral Density
ns	Nanoseconds
OH*	Oxygen-Hydrogen Radical
PMT	Photomultiplier Tube
ps	Picoseconds
QE	Quantum Efficiency
SNR	Signal to Noise Ratios
SpaceDOC	Space Flight Systems Development and Operations Contract
UV	Ultraviolet
VDC	Volts (direct current)